MLFF-DATA Stage and Data-Contract Specification

MLFF-DATA0

mdstats project

2026-08-04 (through DATA9B3A)

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1 Scope

This specification freezes the canonical stage order and controlling data contracts for the future `mdstats.training_data` branch. It is subordinate to `mlff_training_data_architecture.{md,pdf}` and normative where it uses **MUST**, **MUST NOT**, **SHALL**, or **SHALL NOT**.

No public runtime object is implemented at MLFF-DATA0. DATA1 through DATA8, DATA9A hardening through DATA9A9c, and DATA9B1 through DATA9B3A are implemented; later stages MUST NOT expose a public placeholder before its specification, tests, and gate are complete. The mdstats core has no mandatory MACE, PyTorch, or replay-data dependency.

2 Normative principles

1. Source facts, eligibility, partition, selection, weighting, exposure, and acquisition are separate immutable record families.
2. A frame that has contributed a gradient is not independent validation evidence for that model.
3. Cross-validation SHALL train one fresh model per held-out evaluation fold.
4. A held-out evaluation fold SHALL NOT control stopping or checkpoint choice.
5. Every fold SHALL have a checkpoint-monitor domain disjoint from both its gradient-training domain and held-out evaluation fold.
6. Cross-validation SHALL be bound to the complete `TrainingProtocolIdentity` used for final training. Naive and replay protocols are different protocols.
7. Feature fitting, label-derived difficulty features, E0 fitting, and selection SHALL inspect only the applicable training domain.
8. Locked tests SHALL NOT affect fitting, selection, protocol choice, stopping, checkpointing, uncertainty calibration, or acquisition.
9. Locked tests SHALL be absent from training configurations and activated only after a `ProtocolFreezeRecord` and selected committee identity exist.
10. Partition-critical system-profile features SHALL exist before the outer partition is locked.
11. The first MACE adapter SHALL support one target label domain, an optional replay head, and fixed-file native training only.
12. The first MACE adapter SHALL verify native head ordering, checkpoint control, and target/replay exposure realization against its version lock.
13. Replay retention SHALL be governed by an explicit constraint and applied to saved candidate checkpoints.
14. Dynamic per-epoch resampling SHALL require a custom runtime adapter or explicit multi-job protocol; fixed files alone SHALL NOT claim that feature.
15. Active-learning calibration SHALL be bound to the actual final committee and a declared applicability domain.
16. Existing frame roles SHALL be inherited unchanged by active-learning child datasets unless a new evaluation lineage is explicitly created.

3 Record-family contract

Record	Owns	Must not own
`TrainingDataSource`	source path, hashes, composition, controls, ensemble, quality, label domain	frame eligibility or split role
`TrainingFrameRecord`	source-bound frame facts and identity references	eligibility, partition, selection, exposure, acquisition
`FrameEligibilityDecision`	post-DFT label/quality outcome	partition or training membership
`PartitionAssignment`	one statistical role under one partition policy	feature-selection outcome
`PartitionFeasibilityReport`	support for requested roles and declared reductions	fabricated independent cohorts
`PartitionIndependenceReport`	evidence grade, purge, correlation, and duplicate limits	claim of stronger independence than observed
`SelectionAssignment`	selected/not-selected and reasons	epoch use count
`ExposureAssignment`	head, epochs, weights, and use counts	independent-evidence claim
`MaceExposureRealizationRecord`	actual loader counts, duplication, batches, and property exposures	intended exposure only
`AtomicReferenceFitRecord`	training-domain E0 corrections and residual	held-out labels
`TrainingProtocolIdentity`	complete model/data/replay/objective/exposure/checkpoint protocol	mutable runtime observations
`ProtocolFreezeRecord`	frozen selected protocol and committee identities	evaluation results
`CandidateAdmissibilityDecision`	pre-DFT geometry and trajectory safety	DFT convergence claim
`AcquisitionDecision`	calibrated or rank-only acquisition result	post-DFT eligibility
`MaterialProfileIdentity`	user-declared phases, geometry, chemistry modifiers, and optional extensions	structural feature values or automatic classification claims
`AtomGroupCatalog`	immutable group selectors and phase linkage	per-frame group membership arrays unless owned by a provider artifact
`ConditionAxisCatalog`	axes whose coverage may matter	observed coverage or partition assignment
`IndependenceAxisCatalog`	candidate sources of independent evidence	proof that independent realizations exist
`MaterialProfileContracts`	digest-bound aggregate of the four declarative profile families	physical features, events, or validation results

Every serialized record SHALL include a versioned schema and deterministic content digest. Policy and decision records SHALL additionally carry their policy identity and explicit failure reasons where applicable. Identity records SHALL carry their declared parent or provider lineage rather than a fictitious policy digest.

4 Identity contract

4.1 Source occurrence

`frame_uid` SHALL be derived from source identity and source frame index. It identifies an occurrence, not a unique physical configuration.

4.2 Geometry and label identities

The following identities SHALL be separate:

```
geometry_fingerprint
label_payload_digest
labeled_configuration_fingerprint
```

`geometry_fingerprint` SHALL exclude energy, force, and stress labels. `label_payload_digest` SHALL include selected labels and label-domain identity. The combined fingerprint SHALL identify the same geometry with the same labeled payload.

Leakage audits SHALL use exact UID overlap, exact geometry overlap, exact labeled-configuration overlap, near-duplicate geometry/descriptor distance, and forbidden temporal proximity.

5 Label-domain and energy contract

A label-domain fingerprint SHALL be decomposed into:

```
TheoryIdentity
EnergyReferenceIdentity
DerivativeConvention
NumericalQualityProfile
SoftwareProvenance
```

A versioned compatibility policy SHALL classify differences. Exact equality of all provenance fields is not required; theory- or energy-reference differences cannot be silently combined.

The first adapter SHALL export one target label domain per MACE bundle. Multiple incompatible target domains SHALL produce multiple bundles.

The selected VASP energy channel SHALL be named, complete, and consistent with the derivative labels. The channel identity SHALL be preserved in provenance.

6 Atomic-reference contract

The structural `AtomicReferenceIdentifiabilityReport` SHALL record:

```
count matrix
rank
singular values
condition number
null-space dimension
identifiable combinations
policy outcome
transfer limitations
```

It SHALL NOT contain fitted corrections or fit residuals.

Each fold and final training domain SHALL receive a separate `AtomicReferenceFitRecord` containing:

```
training-domain frame UIDs
element support
identifiability-report digest
foundation-checkpoint digest
fitted corrections
fit residual
solver/tolerance
policy outcome
```

The fit SHALL exclude held-out evaluation, checkpoint-monitor, outer monitor, calibration, and locked-test labels. Missing elemental support SHALL fail or invoke an explicit alternative policy.

An emitted MACE configuration SHALL contain the exact version-supported E0 value, normally an explicit atomic-number mapping. A conceptual record name SHALL NOT be written into the `E0s` field.

7 Cell, strain, stress, and virial contract

ASE cell vectors SHALL be treated as rows. Fractional row vectors SHALL map as

$$\mathbf{r}_{\mathrm{row}} = \mathbf{s}_{\mathrm{row}}\mathbf{H}.$$

For reference cell $\mathbf{H}_0$ and current cell $\mathbf{H}_t$, the reported deformation gradient acting on Cartesian column vectors SHALL be

$$\mathbf{F} = \left(\mathbf{H}_0^{-1}\mathbf{H}_t\right)^T.$$

The implementation SHALL record whether internal calculations use an equivalent right-acting row-vector map. Tests SHALL include nonsymmetric shear and rotated stretch fixtures.

Canonical `REF_stress` SHALL be a symmetric Cartesian 3 x 3 Cauchy-stress tensor in eV/Angstrom^3, using the ASE/MACE sign convention verified by the version- locked adapter. Virial and stress SHALL use distinct keys.

The export gate SHALL test units, sign, Voigt order, shear factors, and MACE read-back. Missing stress MAY use `config_stress_weight=0.0` under an explicit heterogeneous-label policy.

8 Event, feature, and blinding contract

Full-resolution event detection SHALL run before ordinary thinning. Protected events include coordination changes, site changes, ring crossings, topology changes, strain extrema, and high but physical restoring-force excursions.

DATA4 SHALL provide partition-critical system-profile features before the outer partition is locked. For LTA these SHALL include resolvable coarse ring-site, off-center, coordination-change, site-change, ring-crossing, and framework- integrity states.

Raw feature providers SHALL be partition-independent. Scaling, PCA, whitening, or fitted metrics SHALL be represented by separate static templates and fitted records:

```
FeatureMetricPolicyTemplate
FoldFeatureTransform
FoldFeatureMetricFit
FinalFeatureTransform
FinalFeatureMetricFit
```

A static template SHALL define feature blocks, scaling rules, block/species weights, retained dimensions, missing-block behavior, distance metric, dtype, and tolerance. A fitted record SHALL contain only parameters learned from its declared training domain.

Foundation descriptors may be computed for all domains. Foundation-model residuals require DFT labels and SHALL be exposed only in a `TrainingDifficultyFeatureCatalog` for the applicable training domain. Outer monitor, calibration, held-out-fold, and locked-test residuals SHALL remain blinded until authorized evaluation.

9 Partition feasibility and outer-role contract

A `PartitionRoleBudgetPolicy` SHALL declare requested roles, minimum block counts, minimum independence grades, and allowable reductions. A `PartitionFeasibilityReport` SHALL classify support before assignment.

Allowed outcomes include:

```
fully_supported
supported_with_temporal_blocks_only
calibration_deferred
challenge_set_external_only
reduced_cross_validation_folds
insufficient_for_locked_test
insufficient_for_requested_roles
```

The workflow SHALL NOT fabricate every role from a short trajectory to satisfy fixed percentages.

Every feasible target label domain may define:

```
development_pool
outer_monitor_validation
uncertainty_calibration
locked_interpolation_test
zero or more locked_challenge_tests
```

The calibration domain MAY be deferred to later independent calculations. Locked tests SHALL remain operationally sealed.

The LTA profile SHALL use hierarchical applicable schemas rather than a global Cartesian product:

```
unstrained: composition x temperature x regime
strained: composition x reference-condition x strain-mode x sign x regime
```

Every cohort SHALL emit a `PartitionIndependenceReport` with machine-readable grades such as independent replica, independent ordering, independent run, purged temporal block, slow-state not decorrelated, or insufficient independence.

10 Cross-validation and training-protocol contract

A `TrainingProtocolIdentity` SHALL bind:

```
foundation checkpoint and head
naive or multi-head mode
replay source, selection, and monitor
training objective and property weights
target/replay head weights
exposure backend and balancing policy
checkpoint metric and checkpoint-control policy
replay-retention policy
optimizer, scheduler, epoch cap, and seed policy
MACE adapter lock
```

A `CrossValidationJobFamily` SHALL contain K independent jobs using the same complete protocol. For fold k:

1. designate a held-out evaluation fold;
2. form the non-evaluation domain;
3. carve a deterministic purged checkpoint-monitor split;
4. fit transform, metric, E0, difficulty features, and selector only on the remaining gradient-training domain;
5. initialize a fresh model and optimizer;
6. train under the declared MACE checkpoint-control policy;
7. select and freeze a checkpoint without inspecting the held-out fold;
8. evaluate only then on the held-out fold;
9. emit out-of-fold predictions and independence grades.

A rotating inner-validation fold inside one evolving model is prohibited. Cross-validation results from a naive protocol SHALL NOT be used as validation of a replay protocol.

11 Selection-budget contract

The selector SHALL construct one deterministic master order with mandatory anchors followed by quota-interleaved evidence classes:

```
representative coverage
species-environment coverage
rare events
```

```
descriptor FPS
difficulty enrichment
```

`SelectionBudgetPolicy` SHALL define counts or fractions and a deterministic deficit-redistribution rule. Later evidence classes SHALL NOT be starved because an earlier selector consumed the size budget.

Near-duplicate pruning SHALL occur during master-order construction. Requested sizes SHALL be exact prefixes. A size smaller than the mandatory-anchor count is infeasible.

Species-specific Li, Na, and K environment selection SHALL be available in the LTA profile. Whole-cell framework-dominated descriptors alone are insufficient.

12 Training objective, weights, and checkpoint metrics

`TrainingObjectivePolicy` SHALL define the loss family, energy/force/stress weights, head weights, normalization, robust-loss settings, and missing-label behavior.

`ConfigurationWeightPolicy` and `PropertyWeightPolicy` SHALL define condition-, regime-, event-, quality-, and property-specific weights. Selection, weighting, and exposure are separate decisions.

The first adapter SHALL use standard MACE configuration/property weights and SHALL NOT claim a species-aware atomwise loss. It SHALL report species-resolved force metrics and impose profile focus-group acceptance or checkpoint constraints. Any custom species-aware loss SHALL create a different `TrainingProtocolIdentity`.

`CheckpointMetricPolicy` SHALL define the primary target scalar plus energy, force, stress, species, worst-condition, rare-event, and replay-retention constraints.

13 Exposure and MACE realization contract

Exposure backends are:

```
NATIVE_MACE_FIXED
CUSTOM_EPOCH_RESAMPLE
MULTI_JOB_RESAMPLE
FINAL_REFIT
```

The first adapter SHALL support only `NATIVE_MACE_FIXED`. Dynamic epoch resampling SHALL NOT be represented by static files alone.

Every run SHALL emit a `MaceExposureRealizationRecord` containing:

```
real_pt_data_ratio_threshold
pre-MACE target/replay counts
post-MACE effective target/replay counts
implicit duplication factor
expected and observed batches
configuration, energy, force, and stress exposures
```

The adapter SHALL disable implicit target duplication where supported. Otherwise it SHALL bind the realized duplication to `TrainingProtocolIdentity` and fail if observed loader counts differ from the accepted plan.

14 MACE checkpoint-control and replay contract

The initial adapter target is `mace-torch==0.3.16`, with a tested compatibility matrix. It SHALL capture package digest, source/tag identity, CLI help, and key parser, loader, and training-loop source digests.

MACE 0.3.16 uses the last validation head for native scheduling, patience, and best-checkpoint decisions. The accepted first mode is:

```
NATIVE_TARGET_LAST_WITH_EXTERNAL_CONSTRAINT_AUDIT
```

The adapter SHALL:

1. verify that the target monitor is the last validation head;
2. prevent replay-head behavior from terminating training;
3. retain every evaluation checkpoint;
4. externally evaluate target, focus-group-resolved, and replay monitors;
5. apply `CheckpointMetricPolicy` deterministically;
6. fail closed when version-tested behavior changes.

Replay preparation SHALL precede replay-aware cross-validation. The replay monitor SHALL be disjoint from replay training. `ReplayRetentionPolicy` SHALL define baseline, metric, tolerated degradation, and failure/override behavior.

15 MACE artifact contract

Extended XYZ SHALL contain only MACE-readable labels, weights, and compact stable identities. Complete provenance and selection reasons SHALL live in a sidecar manifest keyed by `frame_uid`.

The development bundle SHALL contain no locked-test path and SHALL include the complete `TrainingProtocolIdentity`, objective, checkpoint metric, replay plan, checkpoint-control policy, E0 mapping, and adapter lock.

The following artifacts SHALL be separate:

```
development_bundle/
calibration_bundle/
sealed_evaluation_bundle/
evaluation_activation/
evaluation_results/
```

A sealed evaluation bundle MAY be prepared early. An `EvaluationActivationDecision` SHALL require a `ProtocolFreezeRecord`, selected committee identity, complete training-protocol digest, and checkpoint-selection decision.

16 Calibration and active-learning contract

Out-of-fold predictions MAY diagnose uncertainty ranking. Numerical thresholds SHALL be calibrated from the actual final committee on a dedicated calibration cohort.

`CalibrationApplicabilityDomain` SHALL record covered elements, compositions, temperatures, strains, cell sizes, site/event classes, descriptor-distance range, force/stress range, and framework-integrity state.

`CalibrationTransferDecision` SHALL classify candidates as:

```
within_calibrated_domain
rank_only_outside_domain
recalibration_required
rejected_incompatible_domain
```

Locked tests are forbidden from calibration and acquisition.

Pre-DFT candidates receive `CandidateAdmissibilityDecision`. Child datasets SHALL inherit existing roles unchanged by default. Selection-biased new labels enter the development pool; independent random labels may form new validation/calibration cohorts; predeclared challenges may form new locked challenge sets. A full repartition SHALL create a new evaluation lineage.

17 Stage gates

17.1 MLFF-DATA0

Freeze architecture, stage order, data contracts, dependency graph, references, PDFs, tests, and release records. No runtime API.

17.2 MLFF-DATA1 - implemented in 0.20.29a0

Implemented source-independent autocorrelation, complete-frame blocks, purge, and deterministic assignment. Stage 11 public and serialized behavior is preserved by exact parity fixtures.

17.3 MLFF-DATA2 - implemented in 0.20.30a0

Implemented deterministic manifest/discovery, immutable VASP source catalogs, composition reconstruction, ensemble references, named energy policy, decomposed label identities, complete-link label-domain grouping, optional quality/production references, and per-label-domain structural atomic-reference identifiability. No frame eligibility, fitted E0 value, or MACE artifact is introduced.

17.4 MLFF-DATA2A - automatic review-manifest inference gate - implemented in 0.20.60a0

DATA2A is an operational refinement of DATA2 source discovery. The first campaign `prepare` invocation SHALL read completed VASP XML controls and cell matrices, infer reviewable ensemble, thermostat, target-temperature, timestep, and fixed-cell metadata, and propose LTA strain intent from filenames. Filename values at or above one SHALL default to percent semantics, so `hydro+5` means +5%, while `hydro+0.05` remains a fractional +0.05 volume change.

A proposed fixed-cell strain relationship SHALL NOT become an operational assertion or reference group until the actual right-polar stretch and volume ratio match the exact LTA hydrostatic, volume-preserving orthorhombic, or symmetric right-polar shear definition. Candidate references may differ in temperature; temperature ranks geometry-passing candidates but never overrides a failed cell relationship. Rejected or ambiguous candidates SHALL remain diagnostic only, emit a warning, and leave both the strained run and that candidate relationship in the conservative ungrouped state. Manifest approval remains mandatory. The DATA2 source gate SHALL also recover a demonstrably trailing interrupted XML stream when controls, atom identities, and complete ionic records are unambiguous. It SHALL retain complete records, conditionally recover one complete but unclosed final calculation, and record the interruption as soft-quality evidence. Mid-file corruption and critically incomplete streams remain hard failures.

17.5 MLFF-DATA3 - implemented in 0.20.31a0

Implemented immutable frame facts, manifest-bound occurrence UID, geometry/label identities, exact duplicate and restart detection, post-DFT eligibility, temperature conditions, reference-cell resolution, and the normative ASE-row finite-strain convention. No partition, fitted feature, selection, or MACE artifact is introduced.

17.6 MLFF-DATA4 - implemented in 0.20.32a0

Implemented partition-independent raw physical features, lightweight full-resolution LTA ring/site states, event-anchor and protected-window catalogs before thinning, deterministic canonical-JSON feature caching, and `PartitionRoleBudgetPolicy`. The stage records role requests only; DATA5 retains feasibility and assignment ownership. No fitted transform, partition, selection, label-derived evaluation residual, or MACE artifact is introduced.

17.7 MLFF-DATA5 - implemented in 0.20.33a0

Implemented full-resolution eligible-frame partition units, condition-aware autocorrelation blocks, protected-event-window preservation, role-budget feasibility, deterministic outer development/monitor/calibration/locked-test/purge assignments, independence evidence including replicas and declared structural realizations, independent cross-validation folds with nested checkpoint monitors, blinding boundaries, and fail-closed duplicate/event/role/purge leakage audits. The stage introduces no fitted metric, selection, MACE descriptor, or training artifact.

17.8 MLFF-DATA6 - implemented in 0.20.34a0

Implemented profile-aware universal structural and optional extension features, optional checkpoint-bound MACE atomic descriptor sidecars, canonical final-development and fold-training difficulty domains, data-derived species and focus-group residuals, blinded outer/calibration/fold prediction catalogs, sealed locked-test records, and immutable DATA6 orchestration. MACE and PyTorch remain optional lazy dependencies. No fitted transform, E0 fit, weighting, selection, or training artifact is introduced.

17.9 MLFF-DATA7 - implemented in 0.20.35a0

Implemented canonical final/fold training domains, static feature-metric templates, fold-local and final robust/standard fitted metrics, optional deterministic block PCA, domain-local explicit E0 fits with rank/null-space audits, training objective and configuration/property weight policies, focus-group-resolved checkpoint policies, quota-interleaved deterministic master orders, strict nested ladders, and coverage reports. No MACE training or evaluation artifact is introduced.

17.10 MLFF-DATA8 - implemented in 0.20.36a0

Implemented verified minimal extended XYZ plus sidecar export, explicit atomic-number E0 mappings, a version-locked MACE compatibility/source probe, local replay preparation and disjoint monitor artifacts, fixed-file exposure, checkpoint-control policy, complete `TrainingProtocolIdentity`, loader realization dry runs, independent final/fold job bundles, and sealed evaluation artifacts. No MACE process is executed and no locked test is materialized.

17.11 MLFF-DATA9 - integration qualification and protocol execution

17.11.1 MLFF-DATA9A - integration and production qualification

The mdstats-side hardening is implemented in 0.20.37a0, the offline MACE runtime bootstrap in 0.20.38a0, and real MACE artifact realization/execution smokes in 0.20.39a0. Implemented contracts include foundation-residual E0 fitting, portable artifact paths, complete extended-XYZ round-trip validation, replay property/provenance/element audits, explicit selection-level job generation, source-declared dependency manifests, isolated local-artifact installation, installed-runtime qualification, v0.3.16 scalar-literal YAML serialization, fixed-file target/replay weight realization, genuine parser and loader dry runs, one-epoch training, checkpoint/head inventory, target-head extraction, and finite evaluation round trips. The supplied dependency bundle and MPA-0 checkpoint pass these gates. DATA9A3 in 0.20.40a0 closes the target-corpus portion of the production gate: all 27 trajectories and 37,632 complete frames pass DATA3-DATA5 with a passing leakage audit. No expensive protocol training may begin until the exact frozen production-corpus plan, checkpoint-bound DATA6/DATA7, verified DATA8-generation, and replay-corpus artifacts close the full DATA9A gate.

17.11.2 MLFF-DATA9B - protocol-matched execution and freeze

DATA9B remains gated until DATA9A records a dependency-complete genuine MACE runtime, successful naive and replay smoke jobs, identified MPA-0 and replay artifacts, qualified optional profile-extension evidence where declared, checkpoint-bound production DATA6/DATA7 evidence, and executable DATA8 job artifacts. The target-corpus DATA2-DATA5 benchmark is complete.

Execute independent protocol-matched cross-validation and final-development jobs; catalog every candidate checkpoint; apply target, profile focus-group, stress, and replay-retention constraints; generate training-size learning curves; compare naive and replay protocols; train multiple seeds; construct the final committee; extract the target head; and emit the `ProtocolFreezeRecord`.

17.11.3 MLFF-DATA9B1 - campaign and checkpoint control - implemented in 0.20.56a0

Freeze the exact mode/selection-size/seed experiment matrix after a passed full DATA9A gate. Bind every fold and final run to its exact DATA8 job, protocol family, seed variant, monitor artifacts, and `mdstats-mace-train` wrapper. Inventory every saved checkpoint by contained path, epoch, size, and SHA-256. Record externally evaluated target/focus/stress/worst-condition/replay metrics; apply the frozen `CheckpointMetricPolicy`; reject incomplete or threshold- violating candidates; and select the deterministic constrained optimum only when every cataloged checkpoint has a decision.

17.11.4 MLFF-DATA9B2 - execution, aggregation, committee, and freeze - implemented in 0.20.57a0

Supervise exact MACE commands with bounded retries and restart-latest semantics; record immutable process/log/environment evidence after every attempt and terminate complete process groups under the frozen timeout/grace policy; automatically evaluate every checkpoint on target and replay monitors; aggregate protocol-matched folds and seeds; construct learning curves; compare complete naive and replay families; export selected final target heads into a seed committee; emit `ProtocolFreezeRecord`; and activate sealed evaluation only after exact lineage verification.

The runtime and record contracts are complete and pass bounded real-MACE smoke tests. The actual long production campaign remains gated on completing the real production DATA6-DATA8 realization and binding the intended production replay corpora.

17.11.5 MLFF-DATA9B3 - unified campaign CLI and bounded verification - implemented in 0.20.58a0

Expose DATA2-DATA9B2 through one source-checkout UNIX-style command with one TOML configuration, digest-approved manifest, one SQLite orchestration database, compact status/benchmark/result outputs, source-local critical-precision wrapper shims, and fail-closed stage advancement. Wrap production preparation, DATA9A qualification, required real-MACE preflight, restartable training, checkpoint evaluation, protocol comparison, committee export, and protocol freeze. Add bounded committee-wide NVE deployment checks and actionable failure guidance without claiming the analysis-owned RDF/diffusion acceptance responsibilities of DATA11. Qualify replay as a first-class production input: checkpoint-bound pseudo-label provenance, geometry-disjoint train/monitor sets, target-element coverage, and minimum corpus sizes are mandatory unless an exploratory override is explicit.

17.11.6 MLFF-DATA9B3A - cuEquivariance campaign backend - implemented in 0.20.59a0

Freeze the MACE execution backend as `e3nn` or `cueq`; resolve the initialization default once from the active environment; require a real foundation-model CuEq smoke in `doctor`; bind package/CUDA probe evidence to the campaign; and propagate the same backend through DATA6, DATA8, preflight, production training, checkpoint evaluation, and bounded verification. Reject silent fallback and reject `only_cueq=true` for the portable production path.

17.12 MLFF-DATA10

Implement active-learning candidate admissibility, out-of-fold uncertainty diagnostics, final-committee calibration, applicability/transfer decisions, calibrated or rank-only acquisition, burst deduplication, DFT queries, labeled-round ingestion, append-only role inheritance, and immutable child generations.

17.13 MLFF-DATA11

Run complete regression, real-data preparation, partition-feasibility cases, naive and replay protocol-matched cross-validation smoke jobs, MACE checkpoint control and exposure-realization tests, final committee calibration, sealed-test activation, active-learning replay, and performance acceptance.

18 Testing requirements

The branch SHALL include tests for:

- graph acyclicity and forbidden dependencies;
- no edge from locked tests to fit, selection, difficulty, calibration, checkpoint, or acquisition;
- record-family and static-template/fitted-record separation;
- label compatibility classification;
- E0 rank/null-space reporting and exact explicit mapping export;
- fold/final E0 fits excluding monitors and holdouts;
- ASE-row strain convention with nonsymmetric shear and rotated stretch;
- stress units/sign/Voigt/shear round trip;

- exact and near-duplicate geometry across copied/restart sources;
- event-before-thinning ordering;
- partition-role feasibility reductions;
- fold-local transform, metric, E0, difficulty, and selector isolation;
- checkpoint-monitor/held-out evaluation separation;
- complete protocol identity equality between cross-validation and final runs;
- deterministic quota-interleaved master-order prefixes;
- species-resolved checkpoint constraints;
- replay train/monitor disjointness;
- target-last validation-head control for every supported MACE version;
- save-all candidate checkpoint audit;
- implicit replay-ratio duplication detection and realization accounting;
- dynamic-resampling rejection by the fixed-file adapter;
- locked tests absent from development MACE configurations;
- protocol-freeze requirements before evaluation activation;
- final-committee calibration identity and applicability binding;
- append-only role inheritance and immutable dataset lineage;
- clean-wheel import and artifact read-back when runtime APIs are added;
- exact MACE `setup.cfg` dependency-manifest parsing;
- offline local-artifact installation with `--no-index`;
- missing-dependency fail-closed behavior without compatibility stubs;
- genuine CLI help and short-training smoke records once the environment is complete.
- protocol-selected float32/float64 config realization and serialized-model dtype inspection;
- float64-foundation to float32-trained transition evidence and float64 control evidence;

19 Documentation requirements

Each runtime module receives a Markdown/PDF specification before implementation. External algorithms and version-dependent MACE behavior are cited. mdstats-local policy choices, thresholds, and tie-breaking rules are identified as such.

The normative bibliography is maintained in `mlff_training_data_architecture.{md,pdf}`.

19.0.1 MLFF-DATA9A3 - implemented in 0.20.40a0

Qualified the complete bulk-LTA target corpus through DATA5. The target-corpus gate passes with explicit temporal-block and weak-independence warnings. DATA9B remains blocked on the exact frozen production-corpus plan, complete production DATA6/DATA7 realization, verified DATA8 generation, and production replay identity.

19.0.2 MLFF-DATA9A4 - selectable precision implemented in 0.20.41a0

The DATA8 optimizer policy owns the requested MACE fine-tuning precision. Both `float32` and `float64` are supported even when the foundation MPA-0 checkpoint is uniformly float64. Parser realization, execution, saved-model inspection, and target-head inspection MUST agree with the protocol-selected precision. Mixed floating-point state fails closed. Real one-epoch transfer smokes for both precisions are required before release. The production 2,734-frame DATA6–DATA8 realization remains a separate unfinished gate and DATA9B remains closed.

19.0.3 MLFF-DATA9A5a - critical-FP64 execution implemented in 0.20.42a0

Training and inference model bodies remain independently selectable as float32 or float64. For the qualified Python/ASE MACE 0.3.16 path, TF32 is disabled and atomic-energy, virial, and stress reductions plus returned observables are safety-locked to float64. ASE positions, cell, masses, and momenta are audited as float64 before MD. Optimization-time force Jacobians remain in the selected model dtype because the upstream FP32 second-derivative graph rejects an FP64 scalar seed. Production DATA6-DATA8 realization and DATA9B remain separate gates.

19.0.4 MLFF-DATA9A5b - deployment-artifact closure implemented in 0.20.43a0

mdstats owns deterministic export of a uniformly `float32` or `float64` MACE model, exact state conversion, reload validation, immutable provenance, and a model-specific inference smoke. Float32-to-float64 promotion is not precision recovery. LAMMPS is a downstream consumer of the selected model; its internal mixed precision, reductions, Kokkos kernels, and integration behavior are not mdstats implementation or qualification targets. No LAMMPS-specific implementation stage is created.

19.0.5 MLFF-DATA9A6 - observable ownership bridge implemented in 0.20.44a0

The analysis branch now owns a standardized call registry and immutable recipes for implemented structural, topology, dynamical, spectral, and transport observables. The MLFF branch owns only material-profile recommendations, reference/candidate pairing, invocation lineage, and later comparison policy. No RDF, coordination, angle, connectivity, diffusion, VDOS, or conductivity algorithm is duplicated in `training_data`.

19.0.6 MLFF-DATA9A6b - architecture and observable-evidence consistency closure - implemented in 0.20.45a0

The canonical manuals, indices, stage identifiers, and dependency graph are reconciled. `ObservableAnalysisRecipe` rejects self, unknown, forward, and missing dependencies during construction. Capability records expose machine-checkable collection requirements, required/alternative arguments, versioned codecs, owner-signature identities, and result hints. Execution preflights collection semantics and records warnings, runtime versions, and capability digests.

`ObservableCollectionIdentity` records exact scientific collection identity while keeping filesystem paths as non-identity location hints. DATA9A6b introduced the candidate-generation record, runtime evidence, and capability contracts; the remaining verification and leakage gates are closed in DATA9A6c.

The temporary flat enum is renamed `ObservableRecommendationProfile`; The obsolete flat `MaterialValidationProfile` alias has been removed. A separate thermomechanical and energetic validation architecture now owns EOS, elasticity, finite-temperature response, stress-correlation viscosity, phonons, surfaces, interfaces, defects, and migration paths.

19.0.7 MLFF-DATA9A6c - observable evidence and leakage closure - implemented in 0.20.46a0

The bridge SHALL verify caller-supplied collection identities against the actual arrays and reject object-dtype identity inputs. Production evidence SHALL bind both reference and candidate trajectories through symmetric `TrajectoryGenerationIdentity` records whose `output_collection_digest` equals the analyzed collection identity.

Every native analysis result SHALL receive an analysis-owned `ObservableResultIdentity`. The MLFF bridge SHALL store only those identities, paired types, warnings, durations, runtime identity, and upstream lineage; it SHALL NOT define a competing scientific-result schema. A lightweight `MLFFObservableValidationEvidenceRecord` SHALL round-trip and detect tampering without loading result arrays.

Every execution SHALL declare an `ObservableEvidenceRole`. Outer-validation, calibration, locked-test, and external-benchmark roles SHALL bind a comparison policy selected before realized evidence. Locked-test execution SHALL also bind partition policy/assignment, protocol freeze, and explicit evaluation activation. Observable evidence SHALL NOT tune its own comparison policy or retroactively alter selection, training protocol, checkpoint choice, calibration policy, or acquisition.

Runtime evidence SHALL identify the executing mdstats source/version and module hash separately from installed-distribution metadata. Capability identity SHALL include owner implementation source digest and a stable owner-manual ID/URI.

19.0.8 MLFF-DATA9A7a - material-profile and atom-group contracts - implemented in 0.20.47a0

Immutable compositional phase, geometry, chemistry-modifier, structural-extension, atom-group, condition-axis, and independence-axis contracts are public. The runtime-checkable `SystemProfileProvider` owns declarative identity only. DATA4 schema v2 may carry the aggregate contract while retaining v1 read compatibility. The one-phase fallback declares only `all_atoms`; multi-phase and interface systems require explicit atom-group membership. Generic defaults do not activate LTA modules or extensions.

19.0.9 MLFF-DATA9A7b - universal structural selection providers - implemented in 0.20.48a0

The analysis-owned local-structure kernel now supplies chemistry-scaled smooth coordination, nearest-neighbor and radial-basis features, weighted connectivity, local-density proxies, angular Legendre moments, and bond-orientational order. The MLFF provider aggregates these features by declared atom groups and present elements, records generic geometry-only temporal events, and threads immutable catalogs through DATA6 schema v2. DATA7 may fit the `universal_structural` block, and generic per-species environment FPS prefers these descriptors. Existing checkpoint-bound MACE descriptors remain the learned novelty path. DATA6 v1 is read-compatible, generic defaults do not activate LTA, and sealed roles cannot be materialized.

19.0.10 MLFF-DATA9A7c - phase and geometry profiles - implemented in 0.20.49a0

The explicit DATA9A7a material contracts now derive one immutable `PhaseGeometrySelectionPlan`. Crystalline, amorphous, liquid, molecular/gas, and custom phases contribute universal feature-family and generic event-family defaults. Bulk, surface, interface, confined, cluster, and custom geometries contribute atom-group priority ordering and explicit missing-region warnings. An interface composes two or more phase profiles and is not a peer phase value.

DATA6 schema v3 stores the plan, binds the universal structural policy to its digest, filters exposed selection features/events by the enabled families, and prioritizes profile-declared surface/interface/phase groups before generic per-element environment coverage. DATA6-v1/v2 remain readable. The profile also composes advisory IDs for currently executable physical-observable calls without moving any numerical observable into the MLFF branch.

19.0.11 MLFF-DATA9A7d - optional profile-extension and LTA migration - implemented in 0.20.50a0

Canonical DATA4/DATA6 evidence now stores generic partition- and selection-stage `ProfileFeatureCatalog` envelopes. LTA ring, cage, window, site, and crossing payloads remain provider-owned and are consumed only through common frame, atomic-environment, and environment-class adapters. Generic profiles never activate LTA; the explicit `porous_network -> zeolite -> lta` chain is required.

DATA4 schema v3 and DATA6 schema v4 remove LTA-named top-level fields from new serialization while retaining v1-v3 read compatibility and deprecated Python views. Generic feature metrics and selectors derive species from authorized roles, and optional focus groups replace fixed Li/Na/K policies. Structural realization and extension-coverage records replace cation-ordering and LTA-site concepts in generic evidence.

19.0.12 MLFF-DATA9A7e - cross-system qualification - implemented in 0.20.51a0

Run bounded DATA4–DATA7 workflows for generic crystal, amorphous solid, liquid, multiphase interface, and LTA-extension cases. Persist immutable clean-import, per-case, and complete-suite evidence. Generic cases must not import the MLFF LTA implementation modules, activate the `lta` extension, or serialize legacy LTA top-level fields. The LTA case must use the explicit porous/zeolite/LTA hierarchy and generic extension envelopes. Qualification proves software-path generality and lineage, not physical model validity.

19.0.13 MLFF-DATA9A8 - profile-aware observable comparison policies - implemented in 0.20.52a0

Implement frozen rule, threshold, score-uncertainty, comparison-result, and acceptance-decision records over native analysis-owned results. Bind policies to the recipe, observable recommendation profile, optional material-profile contracts, statistical role, atom-group/condition scopes, and pre-execution activation digest. Implement absolute, symmetric-relative, normalized-RMSE, integrated-curve, Jensen–Shannon, peak-shift, and exact-mismatch scores. Reject reverse policy fitting from realized evidence and preserve locked-test leakage gates.

DATA9A8 also removes misleading deprecated public aliases and pre-generalization objective/checkpoint/partition/coverage schemas while retaining only the currently justified DATA4/DATA6 cache readers.

19.0.14 MLFF-DATA9A9a - restartable checkpoint-bound DATA6 model sweep - implemented in 0.20.53a0

Derive the exact DATA5-authorized descriptor and prediction frame sets from the active DATA6 policy. Persist per-frame descriptor and prediction sidecars with file and numerical content digests. Maintain an atomically replaced `Data6ModelSweepCheckpoint` that supports bounded execution, interruption, failure evidence, resume, and corruption recovery. Bind a complete sweep to DATA6 schema v5 and reuse persisted predictions when constructing residual and blinded-summary catalogs. Locked-test, purge, excluded, and otherwise unauthorized frames remain outside the requested union.

19.0.15 MLFF-DATA9A9b - production DATA6–DATA8 materialization - implemented in 0.20.54a0

Freeze one complete DATA6 sweep together with every canonical final and fold DATA7 domain, all fitting/selection policies, the exact replay train and monitor artifacts, the foundation checkpoint, and the MACE execution policies. Persist one verified DATA7 bundle per domain, resume after interruption, invalidate DATA8 after any upstream modification, and emit a verified final/fold DATA8 artifact tree. `ProductionMaterializationRecord` binds the completed DATA7 and DATA8 evidence and is the required input to production qualification and DATA9B.

19.0.16 MLFF-DATA9A9c - production-gate integrity closure - implemented in 0.20.55a0

DATA9A9c binds production qualification to an immutable `ProductionCorpusPlan` containing exact source/frame identities, expected runs, condition summaries, fold count, optional-extension evidence requirements, and required DATA9A artifacts. Foundation descriptors/predictions and foundation-residual E0 are derived from verified DATA6/DATA7 evidence rather than caller Booleans. Replay semantic identity includes numerical energy, force, and stress payloads while excluding intentionally transformed configuration weights. Foundation and replay paths are non-identity location hints. DATA8 is built in a hidden staging tree, promoted to a content-addressed generation, and exposed through an atomic pointer switch. Materialization loaders reverify artifacts before deserialization. DATA9B remains closed until the real production plan passes this gate.

The control layer is implemented and qualified on a bounded complete workflow. The full 2,734-frame corpus must still finish DATA9A9a and then execute this DATA9A9b plan before DATA9B protocol-matched training and freeze begin.

19.0.17 MLFF-PERF1 - profiled campaign execution and bounded-state persistence - implemented in 0.20.62a0

Require one normalized VASP decode per source, bounded source concurrency, checksummed resumable frame caches, constant-time catalog access, vectorized per-frame numerical kernels, and memory-bounded DATA4 persistence. Approval must not implicitly start expensive work. Every long stage and external subprocess must expose actionable progress, elapsed time, and either ETA or heartbeat evidence. Production validation must use the complete supplied LTA corpus and preserve scientific identities across optimized and reference kernels.

19.0.18 MLFF-PERF2 - resource-aware parallel and GPU execution - implemented in 0.20.63a0

The campaign SHALL detect effective CPU threads, currently available RAM, selected CUDA availability, and free VRAM. Automatic plans SHALL target configurable fractional budgets whose defaults are 0.80. Worker assignment SHALL be bounded by independent task count, CPU budget, estimated per-worker memory, and memory reserved for the accumulated parent-side result. Explicit worker overrides SHALL NOT bypass those hard bounds.

Source ingestion, DATA3 construction, raw DATA4 features, and profile-specific trajectory features SHALL support process-level parallel execution. Object-heavy paths SHALL NOT use Python thread pools. High-volume workers SHALL suppress nested BLAS/OpenMP parallelism and release native state after each trajectory. LTA process results SHALL use compact typed columns rather than transferring the immutable record hierarchy. Scientific field values and ordering SHALL match the serial path.

DATA6 SHALL use native MACE graph batching only when the source-locked MACE adapter qualifies that path. Automatic batch size SHALL be bounded by the configured fraction of free VRAM, and CUDA OOM SHALL reduce the batch and retry without invalidating already verified frame artifacts. DATA8, preflight, and training SHALL receive one CPU/RAM-bounded MACE DataLoader worker plan. GPU acceleration SHALL be used only for kernels whose arithmetic intensity justifies transfer and conversion overhead.

All plans and long-running stages SHALL emit informative progress. Production qualification SHALL compare serial and parallel scientific identities and execute the complete supplied LTA corpus, including interrupted XML sources.